Model Predictive Control

Chapter 1b: Organization

Prof. Melanie Zeilinger

ETH Zurich

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Course information

Professor:	Melanie Zeilinger, *mzeilinger@ethz.ch*
Teaching Assistants :	Marcell Bartos, *mbartos@ethz.ch* Sabrina Bodmer, *sabodmer@ethz.ch* Marco Heim, *maheim@ethz.ch*
Lectures:	Thursdays 10.15-12.00, ML D 28
Recitations:	Thursdays 12.15-13.00, ML D 28
Office hours:	Directly after the recitation, ML D 28
Exceptions:	9 April (Easter break), 14 May (Ascension Day)
Lecture Notes:	On Moodle, beginning of the week
Problem Sets:	On Moodle, beginning of the week
Recordings:	Link and password on Moodle

Exam & Grades

Written Exam

- in examination session
- only offered in the session after the course unit (repetition only possible after re-enrolling)

Ungraded Take-home Project (optional)

- understand and apply the lecture material
- alone or in group
- will be published during the semester

Literature (Note: No textbook required)

Model Predictive Control:

- Model Predictive Control: Theory, Computation, and Design, James B. Rawlings, David Q. Mayne & Moritz M. Diehl, 2020 Nob Hill Publishing
- Predictive Control for linear and hybrid systems, F. Borrelli, A. Bemporad, M. Morari, 2017 Cambridge University Press
- Predictive Control with Constraints, Jan Maciejowski, 2000 Prentice Hall

Optimization:

- Convex Optimization, Stephen Boyd and Lieven Vandenberghe, 2004 Cambridge University Press
- Numerical Optimization, Jorge Nocedal and Stephen Wright, 2006 Springer

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Parts of the slides in this lecture are based on or have been extracted from:

- Linear Dynamical Systems, Stephen Boyd, Stanford
- Convex Optimization, Stephen Boyd, Stanford
- Model Predictive Control, Manfred Morari, University of Pennsylvania
- Model Predictive Control, Colin Jones, EPFL
- Model Predictive Control, Francesco Borrelli, Berkeley

Class Schedule (tentative)

Lecture 1 19.02.	Introduction to MPC & Organization, System Theory Basics
Lecture 2 26.02.	Optimal Control of Unconstrained Systems
Lecture 3 05.03.	Introduction to Convex Optimization
Lecture 4 12.03.	Introduction to Convex Optimization & Constrained Optimal Control
Lecture 5 19.03.	Constrained Optimal Control
Lecture 6 26.03.	Invariance
Lecture 7 02.04.	MPC Theory

Class Schedule (tentative)

09.04.	*Easter break*
Lecture 8 16.04.	MPC Practical Considerations & Tracking and Soft-Constrained MPC
Lecture 9 23.04.	Robust MPC I
Lecture 10 30.04.	Robust MPC II
Lecture 11 07.05.	Implementation Methods
14.05.	*Ascension Day*
Lecture 12 21.05.	Advanced Topics
Lecture 13 28.05.	Invited Presentations on MPC Applications